Graph Theory

A graph $G = (V, E)$. The degree of a vertex is the number of incident edges; the handshaking lemma says $\Sigma \deg(v) = 2|E|$, so every graph has an even number of odd-degree vertices.

Paths and circuits

An Euler circuit uses every edge exactly once; a connected graph has one iff every vertex has even degree. A Hamiltonian cycle visits every vertex exactly once; no simple test is known.

Trees

A tree is a connected acyclic graph; with n vertices it has $n - 1$ edges. Spanning trees: Kruskal and Prim find a minimum spanning tree.

Colouring and planarity

The chromatic number $\chi(G)$ is the fewest colours for a proper vertex colouring. Euler's formula for connected planar graphs: $v - e + f = 2$.